

JOYAL JOJO

joyaljojo.connect@gmail.com | 0449959100 | <https://joyal-jojo.github.io/joyal-portfolio/> | Blackburn

PROFILE

Proactive and multidisciplinary Mechanical Engineer with strengths in FEA, product and process optimisation, mechanical design, and root cause analysis. Applies engineering rigour under high pressure, technically demanding environments from vibration analysis to manufacturing optimisation, supported by a Lean Six Sigma Green Belt certification. Demonstrated ability to lead complex projects autonomously, spanning physics informed thermodynamic AI research to hands on prototyping of mechanical systems. Driven by first principles thinking and systems engineering to deliver robust, high performance solutions across diverse engineering contexts.

KEY SKILLS

Technical:

- Lean Six Sigma (Green Belt), Kaizen, DMAIC, Process Mapping, 5S, SMED, VSM.
- Defect prevention & Root Cause Analysis (RCA), FMEA, DFM/DFA, AS1100, AS1210, AS1170 Standards.
- QMS (ISO 9001), Wing Load Validation, Strain Gauge Testing.
- Workflow and Process Optimisation, Forensic Material Analysis, Automation & Control Systems, Rapid prototyping.

Software:

- SolidWorks, AutoCAD, Siemens NX, G-Code/M-Code.
- FEA (Stress/Mesh), PLC Ladder Logic, Metrology, 3D printing.
- Python, C++, MATLAB, Simulink, Antigravity, Claude Code, Gemini and ChatGPT.
- AnyLogic (Discrete Event), Abaqus, ANSYS, Spartan (Linux).
- ERP Basics, Microsoft 365, Google Workspace.

Others:

- Health and Safety Compliance (HSE, WHS, Risk Assessment)
- Adaptable, Fast Learner & Calm under pressure.
- High-pressure stakeholder management, Teamwork & Collaboration.
- Problem Solving & Critical Thinking, Adaptability, Attention to detail.
- Time management and Leadership

Work Rights: Full work rights under Temporary Graduate Visa (Subclass 485) till 10 Mar 2029.

EXPERIENCE

Mechanical Engineer

Mar 2026 - Present

Box Hill Miniature Steam Railway Society, Melbourne, VIC

- Engineering Maintenance & Reliability: Apply core mechanical principles to the troubleshooting and precision repair of steam and diesel-hydraulic locomotives. Manage preventative maintenance schedules to maximize asset uptime and ensure 100% operational readiness of track infrastructure.
- Operational Safety & Risk Management: Execute passenger-carrying operations within a high-traffic, regulated environment. Ensure strict adherence to **OH&S** protocols and technical safety standards, maintaining a zero-incident record during public run days.
- Design Optimization & Structural Repair: Re-engineered track carriage components and structural assemblies using **CAD** to enhance durability and load-bearing capacity. Execute high-precision mechanical fits and modifications to improve fleet reliability.
- Asset Lifecycle Management: Coordinate technical inspections and infrastructure audits, leveraging a systems-thinking approach to minimize lifecycle costs and improve the reliability of miniature fleet assets.

Student Project Engineer | SiRe (Students in Renewable Energy)

Feb 2024 - Nov 2025

The University of Melbourne, VIC

- Multidisciplinary System Design: Led the mechanical design and integration for a 10W small-scale wind turbine prototype, collaborating across electrical, software, and marketing teams to ensure cross-functional system compatibility.
- Rapid Prototyping & Iteration: Demonstrated high adaptability by managing the full project lifecycle from conceptual CAD modeling to physical prototyping, delivering a functional unit that exceeded standard curriculum requirements.
- Technical Leadership: Acted as a key technical liaison, streamlining communication between engineering sub-teams to resolve design bottlenecks and optimise the turbine's structural efficiency.

EDUCATION

The University of Melbourne, Victoria, Australia
Master's in Mechanical Engineering

Feb 2024- Dec 2025

- **Recipient: Capstone Project Mechanical Engineering Merit best project Endeavour Award.**
- **Completed Coursework:** Advanced Control systems, Advanced solid mechanics, Advanced material science, Design and Manufacturing practices, Manufacturing Automation and IT, Industrial simulation, Economic analysis for engineers, Fluid dynamics and Thermodynamics.

National Institute of Technology Calicut, Kerala, India
Bachelor's in Mechanical Engineering

Jul 2019-Apr 2023

- Secured admission into one of India's top engineering institutes by competing against 1M+ candidates in the JEE examination.
- Developed a machine learning model to predict failure pressure in metallic pipelines and researched CFRP based preventive maintenance applications.
- **Completed Coursework:** Engineering optimisation methods, CAD (SolidWorks and AutoCAD), numerical analysis, system modelling and analysis, management and finance.

PROJECT HIGHLIGHTS

Advanced MIMO Control System Design: Designed and implemented a multi-input multi-output (MIMO) controller for a Quanser AERO 2DOF system. Developed a linearized state-space representation and implemented a full-state feedback controller with integral augmentation for precise pitch/yaw tracking. Designed a Luenberger observer for state estimation and validated system performance using MATLAB and Simulink to meet strict steady-state error ($<2\%$) and settling time ($<10s$) requirements.

Design & Finite Element Analysis of a Bicycle Frame: Engineered a high-performance bicycle frame using Autodesk Fusion 360, performing structural analysis for level and inclined cycling conditions. Developed force equilibrium models to determine pedaling and reaction forces. Conducted FEA to evaluate Von Mises and principal stresses, identifying failure risks and implementing design changes to optimize stress distribution. Executed mesh sensitivity analysis and utilized material indices to optimize the strength-to-weight ratio.

Industrial Redesign & DFM/DFA Implementation: Redesigned and fabricated 11 critical 3D printer components to transition from low-volume prototyping to high-volume production using CNC machining, injection moulding, polyurethane casting, and sheet metal fabrication. Applied GD&T and AS1100 standards to produce production-ready drawings. Optimized the assembly line layout using U-shaped cells and Takt time analysis to meet a theoretical demand of 10,000 units per year. Performed comprehensive trade-off studies on fabrication time, scrap rates, and sustainability across multiple production techniques, ultimately recommending cost-effective and scalable manufacturing solutions.

Advanced Materials Forensic Analysis: Performed root-cause analysis of material failure modes (fracture and fatigue) using CPFEM (DAMASK) simulations. Evaluated material suitability for high-stress environments and integrated findings into future design iterations to improve component reliability, longevity, and structural safety.

Pressure Vessel Integrity & Compliance (Viva Energy): Conducted a comprehensive structural integrity assessment of a vertical pressure vessel in association with Viva Energy, Geelong. Performed Finite Element Analysis (FEA) to evaluate hoop and longitudinal stresses under seismic, wind, and hydrostatic loads. Ensured full regulatory compliance with Australian Standards AS1210 and AS1170, calculating remaining operational life and providing retrofitting recommendations to enhance safety margins.

Reverse Engineering of Mechanical : Successfully reverse-engineered a spinal tendon rig actuator (U-block) using SolidWorks using AutoCAD. Produced AS1100 compliant drawings and 3D printed components for integration into a live assembly line. Performed FEA in Abaqus and ANSYS to validate stress distribution and mesh sensitivity for high-precision medical research applications.

Statistical Quality Control & Metrology: Conducted Measurement System Analysis (MSA) and studied calibration techniques for precision instruments including LVDT and Comparators. Applied statistical probability and Machine Learning (Python/C++) to predict failure pressure in metallic pipelines from industrial datasets, uncovering actionable process improvements for manufacturing quality control.

Thermodynamic AI Research & Supercomputing (Capstone): Independently spearheaded a year-long research project developing a physics-informed AI model to predict real gas behavior (Hydrogen) under extreme conditions, earning the Mechanical Engineering Merit Endeavour Award. Leveraged the Spartan Supercomputer (Linux) and Python to process complex thermodynamic datasets. Collaborated with industry mentors from Future Energy Exchange to document change requests and present technical solutions to stakeholders via Microsoft 365 and Google Workspace.

Discrete-Event System Simulation & Optimization: Developed a high-fidelity simulation of a high-volume service environment (HEYTEA) using AnyLogic to identify operational bottlenecks. Analyzed customer arrival patterns and service times using statistical distributions. Applied Lean Six Sigma and manufacturing principles to test improvement strategies, successfully reducing wait times and optimizing resource utilization for enhanced system throughput.

Automated Bio-Environment Incubator and System Validation: Managed the full engineering lifecycle from material selection to hardware assembly, sensor integration, and on-site equipment installation. Developed validation protocols and conducted iterative performance testing to ensure temperature/humidity stability and optimal airflow efficiency. Delivered 100% operational reliability for biological outcomes while ensuring strict compliance with operational safety rules.

Multidisciplinary FOG Mitigation & Systems Thinking: Collaborated with SMEC and a cross-functional team of environmental, mechanical, and chemical engineers to address Fats, Oils, and Greases (FOG) buildup in commercial kitchen infrastructure. Developed an innovative solution focused on behavioral marketing and operational management rather than purely technical filtration, which proved more effective for long-term compliance. This project demonstrated high-level problem-solving flexibility, proving that complex engineering challenges often require integrated, non-technical strategies to achieve sustainable, real-world results.

INTERNSHIP

Engineering Intern (Quality & Overhaul)

May 2022 – Jun 2022

Hindustan Aeronautics Ltd, Bangalore, India

- Supported the Quality Management System (QMS) activities during the overhaul of Dassault Mirage 2000I fighter jets, ensuring strict adherence to aerospace safety standards for mission-critical operational readiness.
- Facilitated **root cause analysis** workshops to investigate fluid system leaks; collaborated with cross-functional engineering teams to define and implement corrective actions to prevent reoccurrence.
- Conducted **process validations** using advanced strain gauge instrumentation and ground vibration testing to verify wing load compliance against design specifications.

Technical Consultant

Aug 2023 – Sept 2023

PricewaterhouseCoopers (PwC), Bangalore, India

- Selected from a highly competitive applicant pool to gain early exposure to enterprise-level technical consulting prior to commencing postgraduate studies.
- Shadowed senior consultants to observe the translation of complex system requirements into actionable commercial strategies.

GHD Group Engineering Consulting Virtual Experience

August 2024

The Forage (Virtual)

- Performed data analysis and options evaluation for rail alignments, summarising findings in a technical memorandum to support design certification plans.
- Analysed stakeholder impacts to ensure project compliance with environmental and societal requirements.

MEMBERSHIPS AND CERTIFICATIONS

Certificates: Lean Six Sigma Green Belt (SSGI), Machine Design (Coursera), Applications in Engineering Mechanics (Coursera), Learning SOLIDWORKS (Udemy), AI for Everyone (DeepLearning.AI), Python Programming (U-Michigan), Business Analytics (LinkedIn), Working with Children Check.

Memberships: Engineers Australia (EA), Mechanical Engineering Student Society (MESS), Students in Renewable Energy (SiRe), Box Hill Miniature Railway Society, Graduate Student Council (GSC).

VOLUNTEER ACTIVITY

President

Mar 2016– Mar 2019

Bala-jana-sakyam (Student body)

- Led a team of 100 student volunteers in organizing community well-being events like higher education awareness programs, Food security initiatives for disadvantaged people etc.